Lab 3: Architecture Justification

Self-Service Coffee Kiosk System | PES1UG24AM193 | PES University, Dept. of CSE

1. Architectural Choice

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

The kiosk is split into a Presentation layer (Kiosk Touchscreen UI), a Business layer (Order Manager and Payment Service) and a Data & Hardware layer (Menu & Pricing Data Store, Receipt Printer Controller, and the external card reader/bank gateway). Each layer only talks to the layer directly below it through the provided and required interfaces shown in the component diagram.

2. Two Reasons for This Choice

- **Clear separation of concerns fits the scenario.** The kiosk has three very different jobs: handling touch-screen interaction, applying ordering rules (3 coffee types, 2 sizes, credit-card-only payment) and talking to hardware and stored data (printer, menu and pricing store). Layering gives each job its own component, so the UI can change (for example a new screen design) without touching the pricing logic, and the printer can be replaced as long as it still provides the IReceiptPrint interface.
- **Right-sized for a single, small, self-contained device.** The kiosk is one machine with a small, fixed menu and one payment method. Microservices would add service discovery, networking and deployment overhead that this scope does not justify, and Client-Server would make the café depend on a central server for something that has to keep selling coffee. Layered keeps the system simple to build, test and maintain, and the Order Manager acts as a single coordinator for the whole order flow.

3. Security Advantage

Payment handling is isolated in one dedicated component, the Payment Service, which is the only part of the system that communicates with the card reader/bank gateway (ICardAuthorisation over TLS/EMV). The Touchscreen UI and Order Manager never see raw card details; the Order Manager only sends an amount through IPaymentProcessing and receives a tokenised success/failure result. Because the UI layer cannot reach the data or hardware layers directly, a fault or compromise in the screen software cannot read card data or tamper with menu prices.

4. Performance Benefit

All layers run on the same kiosk, so calls between them (IOrderEntry, IMenuQuery, IReceiptPrint, IPaymentProcessing) are in-process function or local database calls with no network hops between internal services. Menu and price lookups are small local queries, so the touch-screen responds instantly and the customer can move from selection to payment to printed receipt quickly, even in a busy café. The only network call is the unavoidable card authorisation to the bank.